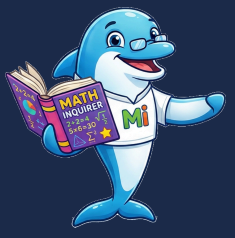
TUTOR GUIDE • LEVEL 1 • TERM 3 • LESSON 8 OF 10

Make a Table



Big idea: A table finds a far-away picture without drawing every one. Draw the first few pictures and put the numbers in a table. Look at how the numbers grow. Then use the pattern to find a number far away.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Number patterns (Primary 1, p. 31) and a growing shape pattern (shape patterns are Primary 2, p. 34) (Number patterns with a regular step, and completing a table of equal groups (Primary 1, p. 31); A growing pattern of shapes made with sticks, and the 10th and 20th picture, is beyond Primary 1; Make a table is a Primary heuristic (p. 13 to 14)) Strand label for the tutor only. NEW TOPIC: the plan has no Primary 1 outcome for a growing shape pattern (shape patterns are Primary 2, p. 34), so a tutor or the owner should confirm the placement. Two school-paper items are close to Start here (a two-row table of equal groups, such as 3 red beads for every 2 blue, and a day-by-day table of pages read): both are Primary 1 and sit at Start here, with new numbers and new contexts. BEYOND PRIMARY 1 ON PURPOSE at Think harder and Go further only: the 10th and 20th picture of a growing stick pattern from a table (the rule double and add 1, said in words), why no row of triangles uses an even number of sticks, and the beyond item Picture 100 (the 100th picture, and working back from a total of sticks, a Primary 4 or 5 pattern idea). Letters for an unknown are Primary 6 (p. 43), so every rule is said in words. Owner decision, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item and a non-routine type (here TAB: a ratio table and a growing-pattern table; PAT: the 10th and 20th term; BACK: work back from a total). Do not describe the lesson as MOE-aligned. Paper strips are used for sticks: never matches or toothpicks.
Objectives	• Build a growing stick pattern, count the sticks, and put the numbers in a table of pictures and sticks. • Say how a table grows (the same number more each time) and use it to find the next numbers. • Complete a two-row table of equal groups, such as 3 red beads for every 2 blue beads. • Think harder: find the 10th and 20th picture from a table, say the rule in words, spot a number that does not fit, and say why a row of triangles never uses an even number of sticks. • Go further: the 100th picture, and working back from a total number of sticks to the number of pictures.
Joining cold	**Remember warm-up covers:** Count in 2s from 4: 4, 6, 8, 10 (spiral: Term 2 Lesson 16, counting in 2s, taught again with a number row); Double a number: 6 and 6, 7 and 7 (spiral: Term 2 Lesson 13, taught again with pictures of two groups); Read a table with two rows: fill in a missing number in a row (taught again with a small table). Only counting to 20, adding two small numbers and the words odd and even (Term 2 Lesson 14; the Mini-lesson says them once). Nothing from the lessons before is needed: the warm-up counts in 2s with a number row, doubles two numbers and fills a missing number in a table of days and stickers. The words table, picture and more each time are taught in Explore and the Mini-lesson. Fairly safe to miss: no later lesson needs the stick pattern, but Mi's Challenge Quest has a gate on it (triangles in a row use 3, 5, 7 sticks; the next number and the sticks for 5 triangles), and Lesson 29 uses a table of two ways to check a sum. A child who misses it watches the recording, does workbook pages 1 and 2 and cuts out the strips if they want to play. A child who joins here cold can follow: the warm-up re-teaches counting in 2s and a small table, Explore is hands-on with strips, and Start here keeps to squares in a row, a bead table and a day table.

Child materials	• Your Term 3 book (Book 6), workbook pages 1 to 5, and the strip page (page 6) with the 14 paper strips already cut out by a grown-up • No printer or no cutting? 14 short strips of paper, or 14 drinking straws or pencils of the same length. Never matches or toothpicks • A pencil, a rubber and spare paper
Tutor materials	• The slide deck • Your own set of strips to show on camera, if you wish • A visible clock No ruler is needed. Nothing needs to be bought. The strips are paper: never matches, toothpicks or anything sharp. Small pieces stay on the table: nothing goes in the mouth or ears; keep them away from younger brothers and sisters. Nobody cuts during the lesson: a grown-up cuts the strips beforehand.
Tiers	Start here (5 questions, pages 1 and 2), Think harder (3, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	Picture 100 at T1, G1, G2: The 10th and 20th picture from a table with the rule double and add 1 (T1), the 100th picture of triangles and of squares (G1), and working back from a total number of sticks to the number of pictures (G2). Finding the 100th term of a pattern from a rule is Primary 4 or 5 pattern thinking; letters for an unknown are Primary 6, so every rule is said in words. Named puzzle: Picture 100. T2 (no row of triangles uses an even number of sticks) is an odd and even reasoning item, also beyond the syllabus. Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: make a table: growing stick pattern and a two-row ratio table; pattern: the 10th and 20th term of a growing pattern from a rule in words; work backwards: from a total number of sticks to the number of pictures; odd or even: why no row of triangles uses an even number of sticks. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first, with your own strips. Check: S1 4, 7, 10; S2 9, 8 and 15; S3 13, 16, 19, 3 and 22; S4 15, 25, 30 and day 5; S5 true, false, true; T1 21, 41 and add 1; T2 No, odd and Yes; T3 4, 13 and 3; G1 201, 301 and 151; G2 15 and 12; G3 10, 20 and 5; I1 6 and 4.
- Tell parents a week before: tear out the strip page (page 6) along the dashed line; a grown-up cuts out the 14 strips and keeps them in an envelope. With no printer: 14 short strips of paper or straws of the same length. Never matches or toothpicks.
- Have the slides open. Have your own strips to show on camera if you wish. Say sticks in the table and strips on the desk, and say the rule in words (double and add 1): never a letter, never the word formula.
- Check the recording captures the tutor and the shared screen. Children video is excluded; any unmuted voice is recorded. Greet the whole group warmly without first names, then say the recording notice; do not use first names while the recording runs. Nobody holds paper up to the camera.
- Be plain with families: completing a table of equal groups is Primary 1 work; the stick pattern, the 10th and 20th picture, working back from a total and the 100th picture go beyond Primary 1 on purpose. The topic is new, so a tutor or the owner should confirm the placement. Do not call the lesson MOE-aligned.
- Slides s-explore3, s-m3 and s-m6 keep the answers off the screen while you ask: the answer is confirmed on the slide after. Read every question, every table and the four options of E2 aloud.

Safeguarding and recording (this lesson)

- The strips are paper and small: they stay on the table, nothing goes in the mouth or ears, and they are kept away from younger brothers and sisters. No matches, toothpicks or anything sharp. Nobody cuts during the lesson; a grown-up cuts the strips beforehand.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a board up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Hopping is quiet and small; a child may stay seated and tap the table.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Make a Table	1:30
2–8	**Remember warm-up**	6	3: Warm-up: count in 2s, Doubles, Read a table	4:52
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:13
12–23	**Explore**	11	3: Explore: build triangles, Explore: guess, then build, Explore: no building now	8:59
23–34	**Mini-lesson**	11	7: Make a table, How does it grow?, Use the table to go on, Why not 3 times?, A rule in words, Your turn, Check: your turn	8:39
34–38	**Brain break**	4	1: Hop in twos	3:17
38–70	**Workbook**	32	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	25:08
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:22
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:49
82–85	**Look-back**	3	1: Look-back	2:23
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:40
	Total	**90**		**67:53**

The script needs 67:53 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 80:25. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 38 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 38, skip in this order: Start with: shorten Explore round 3 (guess the 5th) to one volunteer answer; Then: cut the second item of the Your turn slide (s-m5); Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:30 of 2:00

0:00	SLIDE	Show slide **Make a Table** (s-title)
0:15	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your paper strips. Pause or rewind any time.*
0:26	SAY	Hello, inquirers! Pencil ready, and your paper strips on the table. Cameras can be on or off, as you like.
0:36	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
0:54	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:08	SAY	Today we make a table. A table helps us find a picture that is far away, without drawing every one.
1:18	DO	Greet the whole group warmly. Do not use children's first names on the recording. [0:12]
	NOTE	Keep this to two minutes. Say the recording statement and the microphone routine in the same words every lesson, for anyone who joined late. The routine and the four inquirer rules were introduced in the first lesson of Term 1: do not re-teach them.
	NOTE	Plain statement for families when it comes up: completing a table of equal groups is Primary 1 work; the stick pattern, the 10th and 20th picture, working back from a total and the 100th picture go beyond Primary 1 on purpose, always with the strips in hand. The topic is new, so the placement is for a tutor or the owner to confirm.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. A child without the strips draws sticks as short lines on paper.

2:00 – 8:00 Remember warm-up

script 4:52 of 6:00

2:00	SLIDE	Show slide **Warm-up: count in 2s** (s-rem1)
2:25	REC	*Recording: join in! Count in twos, double two numbers and fill the missing number in the table. Pause if you need longer.*
2:36	SAY	Count in twos from four with me, and clap on every number. Four, six, eight, ten.
2:44	CHILDREN	Everyone counts in 2s from 4 up to 20 together, clapping on every number. [0:20]
3:04	ASK	What comes next? [wait about 12 s for answers]
3:18	ASK	What do we add each time? [wait about 12 s for answers]
3:32	SLIDE	Show slide **Doubles** (s-rem2)
3:58	SAY	Double means the same number twice. Double six is six and six. Double seven is seven and seven.
4:06	CHILDREN	Children show double 6 on their fingers (6 and 6), then double 7. [0:30]
4:36	ASK	What is double six? [wait about 12 s for answers]
4:50	ASK	What is double seven? [wait about 12 s for answers]
5:04	SLIDE	Show slide **Read a table** (s-rem3)
5:30	SAY	Mei sticks two stickers in her book each day. Here is a table. The top row is the day. The bottom row is the stickers.

5:42	**ASK**	How many stickers on day three? [wait about 15 s for answers]
6:00	**ASK**	How do you know? [wait about 20 s for answers]
6:22	**CHILDREN**	Children count on 2 more from 4 with their fingers or strips, and say the numbers aloud. [0:30]
	IF STUCK	If a child is unsure counting in 2s, count along the number row on the slide. For doubles, hold up fingers on both hands. For the table, read across the row: day one, two stickers; day two, four stickers.
	NOTE	Answers: 12, add 2; 12 and 14; 6 stickers (2 more each day). Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 20.

8:00 – 12:00 Sneak-peek puzzles

script 3:13 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:34	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:47	**READ**	Puzzle one. Each box has 4 pencils. Fill in the table. One box: 4 pencils. Two boxes: 8 pencils. Three boxes: blank pencils.
9:00	**CHILDREN**	Children write the number on their own paper. [0:35]
9:35	**READ**	Puzzle two. Picture one has 2 dots. Picture two has 5 dots. Picture three has 8 dots. How many dots does picture five have? I know because blank.
9:50	**CHILDREN**	Children write or draw their answer. [1:15]
11:05	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 12 pencils. Puzzle 2 answer: 14 dots (3 more each time: 11 then 14). Look for good ideas to praise later (a child who wrote the dots for picture 4 first, or who said it grows by 3). Do not write scores or keep a list.

12:00 – 23:00 Explore

script 8:59 of 11:00

12:00	**SLIDE**	Show slide **Explore: build triangles** (s-explore1)
12:25	**REC**	*Recording: pause now. Build triangles in a row with your strips and count the strips each time. Press play when you have tried.*
12:36	**SAY**	Today your strips are the sticks. Make one triangle with three strips. Count them. Now make a second triangle that joins the first one and shares one strip with it, like the picture. Count all your strips. Then a third.
12:56	**CHILDREN**	Children build 1, 2 and 3 triangles in a row, counting all the strips each time, and write the three numbers on their paper. Tutor scans the cameras and the tables. [2:30]
15:26	**ASK**	How many strips for two triangles? For three triangles? [wait about 20 s for answers]
15:51	**SLIDE**	Show slide **Explore: guess, then build** (s-explore2)
16:16	**SAY**	Round two. Guess first. How many strips will you need for four triangles? Then build it and count. Was your guess right?
16:27	**CHILDREN**	Children guess, then build the fourth triangle and count. [1:55]
18:22	**ASK**	What was your guess, and what did you find? [wait about 25 s for answers]
18:52	**SLIDE**	Show slide **Explore: no building now** (s-explore3)
19:16	**SAY**	Round three. Do not build. Use what you have found. How many strips would five triangles need? Tell your grown-up how you know.
19:28	**CHILDREN**	Children think and say a number. [1:00]
20:28	**ASK**	How many strips for five triangles? How did you work it out? [wait about 25 s for answers]

	IF STUCK	A child who counts 3 for each new triangle: ask them to look at the strip the two triangles share. A child who has too few strips draws the sticks as lines. A child who guesses 15: ask what the number for four triangles was.
	NOTE	Main room only; children say what they have made. Tilting the camera to the table is allowed; a face need not be in view; nothing is held up to the camera. Answers: 3, 5, 7, then 9 strips for four triangles; guess 11 for five triangles. Do not say the rule before the children do. Do not confirm 11 until the Mini-lesson.

23:00 – 34:00 Mini-lesson

script 8:39 of 11:00

23:00	**SLIDE**	Show slide **Make a table** (s-m1)
23:25	**REC**	*Recording: keep watching and copy the table. Rewind if you miss a step.*
23:32	**SAY**	Here is a table of what we found. One triangle, three sticks. Two triangles, five. Three triangles, seven. Four triangles, nine. Copy it on your paper.
23:44	**CHILDREN**	Children copy the table onto their own paper. [0:40]
24:24	**ASK**	What goes in the last box, for five triangles? [wait about 15 s for answers]
24:44	**SLIDE**	Show slide **How does it grow?** (s-m2)
25:09	**SAY**	Now look at how the sticks grow. Three to five is two more. Five to seven is two more. The sticks go up by two each time.
25:22	**CHILDREN**	Children trace the plus 2 steps with a finger and say them aloud. [0:20]
25:42	**ASK**	Why only two more, and not three? [wait about 25 s for answers]
26:11	**SAY**	The new triangle shares one stick with the one before it.
26:16	**SAY**	Three, five, seven, nine, eleven: every number is odd. Remember that word, odd.
26:23	**SLIDE**	Show slide **Use the table to go on** (s-m3)
26:48	**SAY**	Use the table to go on. Five triangles, eleven sticks. Two more each time.
26:55	**ASK**	How many sticks for six triangles? [wait about 15 s for answers]
27:13	**ASK**	How many sticks for seven triangles? [wait about 15 s for answers]
27:31	**SLIDE**	Show slide **Why not 3 times?** (s-m4)
27:56	**SAY**	Some people say five triangles need five threes. Look at two triangles. Three sticks and three sticks make six, but only five are there.
28:08	**ASK**	Which stick is counted twice? [wait about 20 s for answers]
28:30	**SAY**	Shared sticks are counted once. The first triangle needs three sticks. Each new triangle needs only two more.
28:40	**SLIDE**	Show slide **A rule in words** (s-m5)
29:04	**SAY**	A rule in words. Double the number of triangles. Then add one. Four triangles: double four is eight, and one more is nine.
29:16	**ASK**	Use the rule for six triangles. What do you get? [wait about 20 s for answers]
29:41	**SAY**	Thirteen. It matches the table. Now we can find any number without drawing.
29:48	**SLIDE**	Show slide **Your turn** (s-m6)
30:12	**SAY**	Your turn. Each jar holds six sweets. Fill in the table for three jars and four jars.
30:21	**CHILDREN**	Children write the two answers on their own paper. Wait 10 seconds before helping. [0:45]
31:06	**SLIDE**	Show slide **Check: your turn** (s-m6a)
31:31	**READ**	Three jars: 18 sweets. Four jars: 24 sweets. Tick your own work with a tick.

	NOTE	Key words to use every time: table, picture, sticks, more each time, rule. Say double and add one in words; never a formula and never a letter. Do not say wrong for a child, only for a number or a table.
	IF STUCK	If a child adds the wrong amount each time, ask: how many more does each new group hold? If a child counts 3 sticks for every triangle, show two joined triangles and ask which stick is shared.

34:00 – 38:00 Brain break

script 3:17 of 4:00

34:00	**SLIDE**	Show slide **Hop in twos** (s-break)
34:25	**REC**	*Recording: stand up and join in! Hop twice for each picture as I call the number.*
34:33	**SAY**	Stand up and get a sip of water. I call a number of pictures. Hop twice for each picture and count in twos. Three pictures is six hops.
34:47	**CHILDREN**	Call 3, 5, 2 and 4 pictures, then 2 and 3 triangles (hop 3, then 2 more). Children hop twice for each picture; a child who sits taps the table twice. [2:30]

38:00 – 70:00 Workbook

script 25:08 of 32:00

38:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
38:25	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
38:34	**READ**	Page 1, Start here. Question 1: squares in a row are made with sticks. Count the sticks for 1 square, 2 squares and 3 squares. Fill in the table. Question 2: for every 3 red beads there are 2 blue beads. Fill in the table. Then: Ali uses 10 blue beads. How many red beads?
39:04	**CHILDREN**	Page 1, questions 1 and 2. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [7:30]
46:34	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
46:59	**READ**	Question 1: 4, 7, 10. Question 2: 9 red beads and 8 blue beads, and 15 red beads for 10 blue beads. Tick your own page with a tick.
47:15	**SAY**	After question 2 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
47:24	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
47:49	**READ**	Page 2. Question 3: squares in a row again. Fill in the table. Then: each new square needs blank more sticks, and 7 squares need blank sticks. Question 4: Ravi reads 5 pages each day. Fill in the table. Question 5: tick true or false for each statement about squares in a row.
48:18	**CHILDREN**	Page 2, questions 3 to 5. [9:00]
57:18	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
57:43	**READ**	Question 3: 13 and 16, then 3 more each time, then 22 sticks for 7 squares. Question 4: 15, 25 and 30. Question 5: true, false, true.
57:57	**SLIDE**	Show slide **Choose your next page** (s-choice)
58:22	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
58:38	**MOVE**	Stand and reach up for 30 seconds, then sit down and choose.
59:08	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [4:00]
	IF STUCK	Q1: count the sticks in one square, then add the sticks of the next square. Q2: add 3 more red and 2 more blue each time. Q3: how many more sticks does each new square need? Q4: add 5 each day. Q5: check each statement against your table.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.

	NOTE	Think harder has the 10th and 20th picture, why a row of triangles never uses an even number of sticks and a wrong number in a table. Go further has Picture 100, working back from a total and the squares against the triangles (beyond Primary 1 on purpose). Do not announce who is on which page. Start here takes a middle child about 18 minutes, so the choice time is short: for most children pages 3 to 5 are for home. Say so plainly, do not skip the answer slides, and take any extra minutes from the flex time.

70:00 – 76:00 Talk time

script 4:22 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:25	REC	*Recording: pause. Tell a grown-up whether the child is right, and test it with your strips.*
70:33	READ	A child says: 5 triangles in a row need 15 sticks, because 5 times 3 is 15. Is the child right? Test it with your strips.
70:47	CHILDREN	Test it with your strips: build 3 triangles in a row and count the sticks. [0:45]
71:32	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:12	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 70 s for answers]
73:28	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
73:53	SAY	Remember the sneak-peek? Puzzle one: twelve pencils. Puzzle two: fourteen dots, because each picture has three more than the one before.
74:04	ASK	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: no. The triangles share sticks. The first triangle needs 3 sticks and each new triangle needs only 2 more, so 5 triangles need 11 sticks. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:49 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:25	REC	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	READ	E1. Each tray holds 7 cups. 1 tray: 7 cups. 2 trays: 14 cups. 3 trays: blank cups. E2. Mei reads 3 pages each day. How many pages does she read in 5 days? Choose the correct answer. Option 1: 8. Option 2: 12. Option 3: 15. Option 4: 35. Write its number in the bracket. E3. Rows of triangles use 3, 5, 7 and 9 sticks for 1, 2, 3 and 4 triangles. Each new triangle needs 2 more sticks. How many sticks for 8 triangles?
77:28	CHILDREN	Children answer E1 to E3 on their own. [2:45]
80:13	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:38	READ	E1: 21 cups. E2: option 3, 15 pages. E3: 17 sticks.
80:44	SAY	Tick your own ticket with a tick. Well done.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:23 of 3:00

82:00	SLIDE	Show slide **Look-back** (s-look)
82:25	REC	*Recording: pause and fill in your look-back box. You can pass.*

82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
83:03	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:40 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**SAY**	Well done, inquirers. Next lesson is Spot the Mistake. Please have the mistake-card page cut out by a grown-up, eight cards, and a pencil. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: squares in a row** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: squares in a row use 4, 7, 10, 13 and 16 sticks. How many sticks for 9 squares?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 28 sticks (3 more each time: 16 and 4 more squares of 3 each is 28). If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Every row of triangles uses an odd number of sticks. Is that always true? Test it with your strips: build 2, 3 and 4 triangles and count.	*Recording: pause. Is it always true that a row of triangles uses an odd number of sticks? Test it with your strips.*	Yes. The sticks are double the triangles, and 1 more. Double a number is even, and 1 more makes it odd: 3, 5, 7, 9.
Make a puzzle for your grown-up	4 (optional)	Make a table that grows by the same number each time. Write the first three numbers. Ask a grown-up for the next number and how they know.	*Recording: pause. Make a table that grows by the same number. Ask a grown-up for the next number and how they know.*	No single answer. For example 4, 8, 12: the next number is 16, because each number is 4 more.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never seen a table. Teach them how a table of triangles finds the number of sticks for 6 triangles.	*Recording: pause. Teach a grown-up how a table finds the sticks for 6 triangles without building them.*	Write 1 triangle 3 sticks, 2 triangles 5 sticks, 3 triangles 7 sticks. It goes up by 2 each time, so 4 is 9, 5 is 11 and 6 is 13.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): Each box has 4 pencils. Fill in the table. 1 box: 4 pencils. 2 boxes: 8 pencils. 3 boxes: ___ pencils.

Answer: 12 pencils

Puzzle 2 (think): Picture 1 has 2 dots. Picture 2 has 5 dots. Picture 3 has 8 dots. How many dots does picture 5 have? I know because ___.

Answer: 14 dots. Each picture has 3 more dots than the one before: picture 4 has 11 and picture 5 has 14.

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Squares in a row are made with sticks. Look at the three pictures. Count the sticks in each.**



(a) 1 square uses ____ sticks.
(b) 2 squares use ____ sticks.
(c) 3 squares use ____ sticks.

Answer: 4; 7; 10

Hint (say this first): Count the sticks in the first square. Then count the sticks you add for the next square.
Worked solution: The first square has 4 sticks. The second square shares one side, so it adds 3 more sticks: 7 in all. The third square adds 3 more: 10 in all.
Common mistakes: Writing 8 for 2 squares (4 + 4): the squares share a side.; Writing 12 for 3 squares.
Time for a middle child: about 3.5 min.

S2 **For every 3 red beads there are 2 blue beads. Fill in the table. The last column is spare.**

Groups	1	2	3	4	5
Red beads	3	6	☐	12	
Blue beads	2	4	6	☐	

Ali uses 10 blue beads. How many red beads does he use? ____

Answer: 9 and 8; 15

Hint (say this first): Each new group has 3 more red beads and 2 more blue beads. Keep going until you reach 10 blue beads.
Worked solution: Red beads go up by 3 each time: 3, 6, 9, 12. Blue beads go up by 2 each time: 2, 4, 6, 8. 10 blue beads is 5 groups, and 5 groups have 15 red beads.
Common mistakes: Adding 3 to the blue beads, or 2 to the red beads.; Writing 10 or 12 for the red beads.
Time for a middle child: about 4 min.

S3 **Siti keeps adding squares to the row. Fill in the table. The last column is spare.**

Squares	1	2	3	4	5	6	7
Sticks	4	7	10	☐	☐	☐	

(a) Each new square needs ____ more sticks.
(b) 7 squares need ____ sticks.

Answer: 13, 16, 19; 3; 22

Hint (say this first): Look at how many sticks you add each time: 4, 7, 10 goes up by the same number. Keep going.
Worked solution: The sticks go up by 3 each time, because each new square shares a side: 4, 7, 10, 13, 16, 19. 7 squares: 19 and 3 more is 22 sticks.
Common mistakes: Adding 4 each time.; Stopping at 19 for 7 squares (that is 6 squares).
Time for a middle child: about 4.5 min.

S4 **Ravi reads 5 pages each day. The table shows the pages he has read in all. Fill in the table.**

Day	1	2	3	4	5	6
Pages	5	10		20		

Ravi has read 25 pages in all. It is day ____ .

Answer: 15, 25, 30; day 5

Hint (say this first): Count on in 5s: 5, 10, 15 ...
Worked solution: Each day adds 5 pages: 5, 10, 15, 20, 25, 30. 25 pages in all is day 5.
Common mistakes: Writing 5 again in each box.; Adding 4 or 6 instead of 5.
Time for a middle child: about 3 min.

True/False statements: Each new square needs 3 more sticks.; 4 squares need 16 sticks.; 6 squares need 19 sticks.

S5 **Look at the squares in a row. Tick True or False for each statement.**

Answer: true; false; true

Hint (say this first): Use your table from question 3. Find each number of squares in it.
Worked solution: Each new square shares a side, so it adds 3 sticks: True. 4 squares need 13 sticks, not 16: False. 6 squares need 19 sticks: True.
Common mistakes: Saying 4 squares need 4 times 4 = 16 sticks.; Not checking the statement against the table.
Time for a middle child: about 3 min.

Think harder (page 3): the taught idea in a new setting

T1 **Mi makes triangles in a row with sticks. Fill in the table. Then finish the rule.**

Triangles	1	2	3	4	10	20
Sticks	3	5	7	9		

Double the number of triangles, then add ____ .

Answer: 21 and 41; add 1

Hint (say this first): The sticks go up by 2 for each new triangle. How many new triangles from 4 to 10? Or double the number of triangles and see how many more you need.
Worked solution: From 4 to 10 triangles there are 6 more triangles, and each adds 2 sticks: 9 and 12 make 21. Or double 10 is 20, and 1 more is 21. For 20 triangles: double 20 is 40, and 1 more is 41. The rule is: double the number of triangles, then add 1.
Common mistakes: Writing 30 or 60 (3 sticks for every triangle).; Writing 11 for the 10th picture (counting on only one step).
Time for a middle child: about 5 min.

T2 **Look at rows of triangles made with sticks.**

(a) Can exactly 20 sticks make a row of triangles with none left over? Circle Yes or No.
(b) Are the sticks in a row of triangles always odd, or always even? Circle one.
(c) Can exactly 25 sticks make a row of triangles with none left over? Circle Yes or No.

Answer: No; odd; Yes

Hint (say this first): Look at 3, 5, 7, 9 and 11 sticks. Are they odd or even?
Worked solution: Each row has double the triangles and then 1 more, so the number of sticks is an even number plus 1, which is odd. 20 is even, so no row of triangles uses exactly 20 sticks. 25 is odd: 25 take away 1 is 24, and half of 24 is 12, so 12 triangles use exactly 25 sticks.
Common mistakes: Saying Yes after finding a close number such as 21.; Thinking that every number of sticks is possible.
Time for a middle child: about 4 min.

T3 **A child made a table for squares in a row. One number is wrong.**

Squares	1	2	3	4	5	6
Sticks	4	7	10	12	16	19

(a) Which number of squares has the wrong number of sticks? Circle 3, 4 or 5.
(b) The right number is ____ .
(c) The sticks go up by ____ each time.

Answer: 4; 13; 3

Hint (say this first): Look at how much the numbers go up each time. Which step is different?
Worked solution: The numbers go up by 3 each time: 4, 7, 10, then 13, 16, 19. The number for 4 squares is 12, but it should be 13.
Common mistakes: Choosing 5 because 16 comes after the wrong number.; Writing 11 or 14.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **Picture 100. Look at the tables for rows of triangles and rows of squares.**

Pictures	1	2	3	4
Triangle sticks	3	5	7	9
Square sticks	4	7	10	13

(a) 100 triangles in a row need ____ sticks.
(b) 100 squares in a row need ____ sticks.
(c) 50 squares in a row need ____ sticks.

Answer: 201; 301; 151

Hint (say this first): Triangles: double the number, then add 1. For squares, how many sticks does each new square add? How many sticks does the first square have?
Worked solution: Triangles: double 100 is 200, and 1 more is 201. Squares: the first square has 4 sticks and each of the other 99 squares adds 3: 99 times 3 is 297, and 4 more is 301. Or 100 times 3 is 300, and 1 more is 301. 50 squares: 50 times 3 is 150, and 1 more is 151.
Common mistakes: Writing 300 or 400 (3 or 4 sticks for every picture).; Writing 200 for the triangles and forgetting the 1.
Time for a middle child: about 6 min.

G2 **Work back from the number of sticks.**

(a) 31 sticks make a row of triangles, with none left over. How many triangles? ____
(b) 37 sticks make a row of squares, with none left over. How many squares? ____

Answer: 15; 12

Hint (say this first): Triangles: take away 1, then halve. Squares: take away 1, then share into groups of 3.
Worked solution: Triangles: 31 minus 1 is 30, and half of 30 is 15. Squares: 37 minus 1 is 36, and 36 shared into groups of 3 is 12.
Common mistakes: Dividing 31 by 3 for the triangles.; Forgetting to take away the 1 first.
Time for a middle child: about 5 min.

G3 **A row of squares and a row of triangles have the same number of pictures.**

(a) With 10 pictures each, the squares need ____ more sticks.
(b) With 20 pictures each, the squares need ____ more sticks.
(c) With 5 pictures each, the squares need ____ more sticks.

Answer: 10; 20; 5

Hint (say this first): Find the sticks for 10 squares and for 10 triangles, then take away. Use the tables on this page.
Worked solution: 10 squares need 31 sticks and 10 triangles need 21 sticks: 10 more. 20 squares need 61 sticks and 20 triangles need 41 sticks: 20 more. 5 squares need 16 sticks and 5 triangles need 11 sticks: 5 more. Each new picture needs 1 more stick for a square than for a triangle.
Common mistakes: Comparing the first pictures only (4 and 3).; Subtracting 10 and 20 from each other.
Time for a middle child: about 5 min.

Investigation (page 5): Strips left over

Easy first step: Make 3 triangles in a row. How many strips do you use? ____
Then: You have 14 strips. What is the most triangles in a row you can make? What is the most squares? Can you use all 14 strips?

Model answer: First: 7 strips. The most triangles is 6 (13 strips, 1 left over). The most squares is 4 (13 strips, 1 left over). No: a row of triangles uses an odd number of strips, so 14 never works, and the squares use 4, 7, 10, 13 or 16 strips, so 14 does not work either.

Look for: A child who builds with the strips and counts; a child who uses the table to find how far 14 goes; a child who says why 14 never fits (odd numbers for triangles; 13 then 16 for squares). Credit any clear reason. Do not mark a child down for stopping at one shape.
Common mistakes: Saying 4 triangles (counting 3 strips each: 12).; Saying 14 strips make 5 squares (5 squares need 16).

Talk and explain (minutes 70 to 76)

A child says: 5 triangles in a row need 15 sticks, because 5 times 3 is 15. Is the child right? Test it with your strips.

Sentence starter: I think ... because ...

> **Model answer: No. The triangles share sticks. The first triangle needs 3 sticks and each new triangle needs only 2 more, so 5 triangles need 11 sticks. We can test it by building 3 triangles: they need 7 sticks, not 9.**

Look for: A child who tests it with the strips, or says the triangles share sticks, or says each new triangle adds 2. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Each tray holds 7 cups. 1 tray has 7 cups. 2 trays have 14 cups.**

3 trays have ____ cups.

> **Answer: 21**

Worked solution: Each tray adds 7 cups: 7, 14, 21. 3 trays have 21 cups.
Common mistakes: Writing 20 or 28.; Writing 3 (the number of trays).

E2 **Mei reads 3 pages each day. How many pages does she read in 5 days? Choose the correct answer. Write its number in the bracket.**

> **Answer: (3) 15**

Worked solution: Each day adds 3 pages: 3, 6, 9, 12, 15. In 5 days she reads 15 pages.
Common mistakes: Choosing (1) 8, which adds 3 and 5.; Choosing (4) 35.

E3 **Rows of triangles use 3, 5, 7 and 9 sticks for 1, 2, 3 and 4 triangles. Each new triangle needs 2 more sticks.**

8 triangles need ____ sticks.

> **Answer: 17**

Worked solution: Go on in twos: 4 triangles 9, 5 triangles 11, 6 triangles 13, 7 triangles 15, 8 triangles 17. Or double 8 is 16 and 1 more is 17.
Common mistakes: Writing 24 (8 times 3): the shared sticks are counted twice.; Stopping early at 15 (7 triangles) or 16.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (counting the sticks; finding the number more each time; using the table; something else); *What helped me* (my strips; the table; checking my answer; I was not stuck); *Next time I will try* (using my strips; checking the table; checking my answer; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Each box has 5 pencils. 1 box has 5 pencils. 2 boxes have 10 pencils. 3 boxes have ____ pencils. 4 boxes have ____ pencils.	15 and 20
H2	Start here	For every 2 pens there are 3 pencils. 2 pens, 3 pencils. 4 pens, 6 pencils. 6 pens, 9 pencils. 8 pens, ____ pencils.	12
H3	Think harder	Triangles in a row use 3 sticks for 1 triangle, 5 sticks for 2 triangles and 7 sticks for 3 triangles. How many sticks for 15 triangles? ____	31
H4	Go further	Squares in a row use 4 sticks for 1 square, 7 sticks for 2 squares and 10 sticks for 3 squares. A row uses 40 sticks. How many squares? ____	13
H5	Start here	Fill in the missing number: 27 + 16 = ____ + 9	34

Online practice (`out/l28/practice`, key `t3l28`): 11 items; 8 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to find a pattern in a table and use it to go further. This week: sticks in rows of triangles and squares, and tables of equal groups.

Term line (plan, section 5): Your child is learning to give a reason for each step. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Make a Table** s-title	TODAY WE WILL: / Make a table to find a picture far away.	Lesson 8 of Term 3, 90 minutes. Objective: build a growing stick pattern, count the sticks, put the numbers in a table, say how the table grows and use it to go further. Plain statement for families: completing a table of equal groups is Primary 1 work; the stick pattern, the 10th and 20th picture, working back from a total and the 100th picture go past the school Primary 1 syllabus on purpose. This is a new topic, so a tutor or the owner should confirm the placement. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 23, mini-lesson 23 to 34, brain break 34 to 38, workbook 38 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice in one line: you and the screen only; a child may keep the camera off.
2	2–8	**Warm-up: count in 2s** s-rem1	Clap on every number. / 4 6 8 10 ? / Count in 2s from 4. What comes next?	Count in 2s from 4 with a clap on each number: 4, 6, 8, 10, 12. This is Term 2 Lesson 16 taught again, with the number row so that a child who has not counted in 2s hears it before saying it. Ask: what comes next (12). Then ask: what are we adding each time (2). The slide never shows the answer. No writing, no marks.
3	2–8	**Doubles** s-rem2	Double means the same number twice. / Double 6 is 6 + 6. / Double 7 is 7 + 7. / What are double 6 and double 7?	Term 2 Lesson 13 taught again. Double 6 is 6 and 6, which is 12; double 7 is 7 and 7, which is 14. A child who does not know holds up 6 fingers on each hand for 6 and 6, and counts. Ask: how do you know (6 and 6 is 12). The answers are not on the slide. This is the pairs idea that the rule double and add 1 uses later. No writing.
4	2–8	**Read a table** s-rem3	Mei sticks 2 stickers in her book each day. / [picture: table] / How many stickers on day 3?	A table has a row of days and a row of stickers. Day 1 has 2 stickers and day 2 has 4. Ask: how many stickers on day 3 (6). Ask: how do you know (2 more each day). Say: a table is a neat way to write numbers in rows. The slide shows an empty box for the answer. A child who is unsure counts on 2 more from 4 on fingers. Joining cold: the table teaches itself.
5	8–12	**Sneak-peek puzzles** s-sneak	PUZZLE 1: anyone can try: / Each box has 4 pencils. Fill in the table. / [picture: table] / PUZZLE 2: think: / Picture 1 has 2 dots. Picture 2 has 5 dots. Picture 3 has 8 dots. How many dots in picture 5? I know because ___.	Puzzle 1 (4, 8, then 12 pencils) is an access question: any child can add 4 again. Puzzle 2 is a thinking question on the lesson idea: each picture has 3 more dots, so picture 4 has 11 and picture 5 has 14; a child who writes 11 has found the next one but not the fifth. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: 12; 14. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.
6	12–23	**Explore: build triangles** s-explore1	Make 1 triangle with 3 strips. / Count the strips. / Make a second triangle that joins the first and shares one strip. Count all your strips. / Make a third. Count all your strips. / [picture: stick_pattern]	Round 1, about 4 minutes. Children build with paper strips on the table: one triangle uses 3 strips. Say aloud that the second triangle shares one strip with the first. A second triangle joined to the first shares one strip, so only 2 more strips are needed: 5 strips in all. A third triangle needs 2 more: 7 strips. Children say each total aloud. The picture on the slide shows two joined triangles only to show what joined means; it never shows a count. Do not write the numbers on the slide: children write them on their own paper as a list. A child who counts 3 more each time is asked to look at the shared strip. A child without strips draws the sticks as lines.

7	12–23	**Explore: guess, then build** s-explore2	You have 1, 2 and 3 triangles in a row. / Guess: how many strips for 4 triangles? / Build 4 triangles. Count. Was your guess right? / A guess is a great start.	Round 2, about 3 minutes, main room. Children guess first (take volunteers: say your guess and why), then build the fourth triangle and count: 9 strips. Ask: how many more strips did the fourth triangle need (2); how many more did the third need (2). Do not say the rule before the children do. Take a different guess without comment: 12 (3 for each triangle) and 9 are both worth hearing. Strips used: 9 of the 14 in the kit. A child who has run out of strips draws the last triangle.
8	12–23	**Explore: no building now** s-explore3	Do not build. / How many strips would 5 triangles need? / Use what you have found. Tell your grown-up how you know.	Round 3, about 3 minutes. Children do not build: they use what they found (3, 5, 7, 9, always 2 more) to guess for 5 triangles: 11. Ask: how many strips for 5 triangles; how did you work it out. Some will say 15 (5 times 3): do not say it is wrong; ask them to look back at the number for 4 triangles. Confirm 11 on the next slide, not here. The answer is not on the slide.
9	23–34	**Make a table** s-m1	[picture: table] / Write what you found in a table.	Show the table that children found in Explore: 1 triangle 3 sticks, 2 triangles 5, 3 triangles 7, 4 triangles 9, and an empty box for 5 triangles. We call the strips sticks in a table: the strips are our sticks. Say: a table keeps the numbers in order so that we can see how they grow. Children write the same table on their own paper. Ask: what goes in the last box (11, as we guessed). Confirm it: this is the answer to Explore round 3.
10	23–34	**How does it grow?** s-m2	[picture: table] / +2 +2 +2 +2 / The same number more each time.	Look at how the sticks grow: 3 to 5 is 2 more, 5 to 7 is 2 more, 7 to 9 is 2 more, 9 to 11 is 2 more. The sticks grow by 2 each time. Ask: how many more sticks each time (2); why only 2 and not 3 (the new triangle shares one stick with the one before). Children trace the plus 2 arrows with a finger. The number of pictures goes up by 1, and the number of sticks goes up by 2. Say once: 3, 5, 7, 9, 11 are odd numbers, every time (odd and even are from Term 2 Lesson 14); this is used again at Think harder.
11	23–34	**Use the table to go on** s-m3	[picture: table] / 2 more each time. What are the next two numbers?	Use the table without building. Start at 11 sticks for 5 triangles and add 2: 13 sticks for 6 triangles, then 15 sticks for 7 triangles. Ask first: how many sticks for 6 triangles; for 7 triangles. Children say the numbers aloud and write them on their paper. The empty boxes show the answers only after they have tried: the next slide does not repeat them. The numbers 13 and 15 are not on the slide.
12	23–34	**Why not 3 times?** s-m4	[picture: stick_pattern] / 3 + 3 = 6?: / Two triangles need 5 sticks, not 6. / They share 1 stick. / Shared sticks are counted once.	The usual wrong idea: 5 triangles need 5 times 3 sticks, which is 15. That counts the shared sticks twice. Show the picture of 2 triangles: 3 sticks and 3 sticks make 6, but the middle stick is shared, so only 5 sticks are really there. Ask: which stick is counted twice (the middle one). Say: the first triangle needs 3 sticks, and each new triangle needs only 2 more. Children who guessed 15 in Explore round 3 are welcome to say so; no name is used.
13	23–34	**A rule in words** s-m5	THE RULE: / Double the number of triangles. / Then add 1. / [picture: table]	A rule in words, with no letters. Look at the table: double 1 is 2, and 2 and 1 more is 3; double 2 is 4, and 1 more is 5; double 3 is 6, and 1 more is 7; double 4 is 8, and 1 more is 9. So the sticks are double the triangles, and 1 more. Ask: how many sticks for 6 triangles using the rule (double 6 is 12, and 1 is 13); does it match the table (yes). Say: now we can find any number without drawing. Do not use the word formula; say the rule in words.
14	23–34	**Your turn** s-m6	TABLE: / Each jar holds 6 sweets. / [picture: table] / Fill in both boxes on your paper.	Two boxes on the children's own paper: a table of equal groups, 6 sweets in each jar: 3 jars are 18 sweets and 4 jars are 24 sweets. Walk the cameras and wait 10 seconds before helping. Nobody holds paper up. The next slide shows the answers and children tick their own work. Answers: 18; 24.
15	23–34	**Check: your turn** s-m6a	Jars 3: 18 sweets. Jars 4: 24 sweets.	Read both answers aloud so a child watching the recording can tick their own work. 6 more each time: 12, 18, 24. Never ask a child to show a wrong answer to the group. A child who writes 16 for 3 jars has added 4: ask how many more each jar holds.

16	34–38	**Hop in twos** s-break	Stand up. I call a number of pictures. / Hop twice for each picture and count in 2s. / Sip some water.	Four minutes. Stand up on camera with a little clear space. Call a number of pictures (3, 5, 2, 4) and children hop twice for each picture, counting aloud in 2s: for 4 pictures that is 8 hops. Then call the number of triangles (2, 3) and hop the sticks: 3 hops, then 2 more. No equipment, no running; a child may stay seated and tap the table twice for each picture. Do not give prizes or compare children.
17	38–70	**Workbook page 1: Start here** s-ws1	Questions 1 and 2. Take your time. / questions S1, S2 / [picture: table]	Read questions 1 and 2 aloud. Children work on their own with their strips; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 2 any child may choose to carry on or try Think harder. Question 1: count the sticks in squares in a row and fill in the table. Question 2: a table of red and blue beads: for every 3 red beads there are 2 blue beads. The table for question 2 is repeated large under the questions so it can be read from the screen.
18	38–70	**Check page 1** s-ws1ans	answers S1, S2	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 4, 7, 10; Q2 9 red beads and 8 blue beads, and 15 red beads for 10 blue beads.
19	38–70	**Workbook page 2: Start here** s-ws2	Questions 3 to 5. Take your time. / questions S3, S4, S5 / [picture: table] / [picture: table]	Read questions 3 to 5 aloud. Question 3: continue the squares table, say how many more each time, and find 7 squares. Question 4: pages read each day, a table for 6 days. Question 5: tick True or False for three statements about the squares. Children may use their strips to check. Wait 10 seconds before helping. The tables for questions 3 and 4 are repeated large under the questions so they can be read from the screen.
20	38–70	**Check page 2** s-ws2ans	answers S3, S4, S5	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q3 13 and 16 in the table, 3 more each time, 22 sticks for 7 squares; Q4 15, 25 and 30; Q5 true, false, true.
21	38–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | The 10th and 20th picture, odd sticks, a wrong number / Go further (page 4) | Picture 100, and working back from a total / Investigation (page 5) | Strips left over / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
22	70–76	**Talk and explain** s-talk	A CHILD SAYS: / 5 triangles in a row need 15 sticks, because 5 times 3 is 15. / Is the child right? Test it with your strips. / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Children test it with their strips: build 3 triangles in a row and count (7, not 9). Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no. The triangles share sticks. The first triangle needs 3 sticks and each new one needs only 2 more, so 5 triangles need 11 sticks. Repeat each child answer in your own words for the recording. The child who is wrong is a child in another class: no cast name is given the wrong answer.
23	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: 12 pencils. Puzzle 2: 14 dots. Each picture has 3 more than the one before.	Reveal both puzzles now. Say: many of you wrote a good reason. Do not read out anyone earlier answer. Puzzle 1: 4, 8, then 12 pencils. Puzzle 2: 2, 5, 8, 11, 14: picture 4 has 11 dots and picture 5 has 14. It was the same idea as the stick pattern: find how much it grows each time, then go on.
24	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, and the four options of E2. Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.

25	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: 21 cups; option 3, 15 pages; 17 sticks. Do not read out any child result.
26	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
27	85–90	**Well done, inquirers!** s-end	Show a grown-up one table you made today. / NEXT LESSON: / Spot the Mistake / Have ready: the mistake-card page, cut out by a grown-up (8 cards).	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson needs the mistake-card page, with the 8 cards cut out by a grown-up (the mat is not cut), and a pencil. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.
28	optional	**Go further: squares in a row** s-opt	Squares in a row: 1 square uses 4 sticks, 2 squares use 7, 3 squares use 10, 4 squares use 13, 5 squares use 16. / [picture: table] / How many sticks for 9 squares?	Optional, only if you are on time. A table of squares in a row: 4, 7, 10, 13, 16, going up by 3 each time (one side is shared). Nine squares: 16 and 4 more squares of 3 sticks each is 16 + 12 = 28; or 3 times 9 is 27 and 1 more is 28. Take two answers; ask: how many more sticks does each new square need (3, one side is shared). Then ask: why 3 and not 4. This is the same move as the triangles. Take the idea, not the speed.
29	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Every row of triangles has an odd number of sticks. Test it. / Make a puzzle | Make a table. Ask a grown-up for the next number. / Teach it back | Show a toy how a table helps. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your paper strips. Pause or rewind any time.*	Gather your workbook, strips, pencil and paper. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Count in twos, double two numbers and fill the missing number in the table. Pause if you need longer.*	Play along. Say each answer out loud before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Build triangles in a row with your strips and count the strips each time. Press play when you have tried.*	Pause. Build 1, 2 and 3 triangles in a row. Count the strips. Guess for 4, then build to check.
23:00	Mini-lesson	*Recording: keep watching and copy the table. Rewind if you miss a step.*	Watch and copy the table. Rewind any step you missed.

34:00	Brain break	*Recording: stand up and join in! Hop twice for each picture as I call the number.*	Stand up. Hop twice for each picture that is called.
38:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up whether the child is right, and test it with your strips.*	Pause. Explain your answer to someone at home, and test it with your strips.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one table you made.*	Show a grown-up your workbook and one table you made.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:30	1:42	ok
remember	6:00	4:52	5:46	ok
sneak	4:00	3:13	3:46	ok
explore	11:00	8:59	10:41	ok
mini	11:00	8:39	10:12	ok
break	4:00	3:17	3:54	ok
work	32:00	25:08	29:58	ok
talk	6:00	4:22	5:11	ok
exit	6:00	4:49	5:39	ok
look	3:00	2:23	2:48	ok
flex	5:00	0:40	0:47	ok
Total	90:00	67:53	80:25	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 5 questions; a middle child needs about 7.5 min for page 1 and 10.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.